

PVLoad-R1 Bench Card

Programmable resistive load for PV I–V characterisation · connections, power order, wiring

1 Arduino Uno → board 1×10 header

Pin	Arduino	Function
1	5V	Logic supply and relay coil supply (~30 mA total)
2	GND	Common reference
3	D6	K1 drive — bypasses R1 (470 kΩ)
4	D7	K2 drive — SHORT, bypasses the whole load
5	D8	K3 drive — bypasses DigiPot U2
6	D9	U2 chip select (CS#)
7	D10	U1 chip select (CS#)
8	D11	SPI MOSI → SDI on both chips
9	D12	SPI MISO ← SDO on both chips
10	D13	SPI SCK → both chips

Header pin 1 is the square pad on the silkscreen. Order down the header is 5V, GND, D6, D7, D8, D9, D10, D11, D12, D13. On the Uno, 5V and GND are on the power header; D8–D13 are in the digital group nearest the USB socket, D6 and D7 in the other group.

2 Banana jacks

Jack	Colour	Connect to
J1	Red	PV+ — cell positive (or the handheld's VΩ lead during bring-up)
J2	Red	+24 V from the bench supply
J3	Black	GND — supply return <i>and</i> cell negative. Shared.

J3 carries two or three things at once. Use a stacking banana plug; a jack with leads jammed alongside each other reads tens of ohms and drifts.

3 Instrument cables and what to set in software

Instrument	Cable to the PC	Set in Part 1 of PVLoad_Main.m
Arduino Uno	USB-A to USB-B (printer cable), straight to the laptop.	SERIAL_PORT = "COM4" List them with serialportlist("available")
Keithley 617 ×2	IEEE-488 (GPIB) only — no USB, no serial. Needs a USB-GPIB adapter ; match it to the installed VISA, which is NI-VISA, so use NI hardware (e.g. GPIB-USB-HS). One adapter serves both: GPIB connectors stack, so run an IEEE-488 cable adapter → meter 1, and a second cable meter 1 → meter 2.	DMM_ENABLED = true DMM_V_ADDRESS = "GPIB0::22::INSTR" DMM_I_ADDRESS = "GPIB0::23::INSTR" Each meter needs its own primary address, set from its front panel (IEEE program; default is 27). Confirm with visadevlist.
617 inputs	Front-panel triaxial INPUT, through a triax-to-dual-banana adapter. HI is the centre	DMM_V_RANGE = 20 (V) DMM_I_RANGE = 0 → picked per level from

	conductor, LO the inner shield.	ISC_FULL
Thorlabs EDFA100P	Plain USB cable. It enumerates as a virtual COM port — no adapter needed. 115200 8N1, CR terminator, all already in the file.	EDFA_ENABLED = true EDFA_PORT = "COM5" Rear interlock must be shorted or it will not enable.

4 Which mode to run

Setup	RUN =	Purpose
Handheld only	"verify"	Seven holds; checks both chips, all three relays, both ladders and R1
Handheld only	"ramp"	Walks the sweep slowly enough to follow on a handheld
Two 617s	"meters"	Identify both meters, configure, take ten readings
EDFA only	"edfa"	Identify, report temperature, ramp the pump
Everything	"sweep"	Full experiment, written to CSV

Each mode opens only what it needs. "board", "ramp", "verify" and "k3" touch no laser and no meters; "edfa" touches no board; "meters" touches neither.

Power sequence

Applies to every configuration. The cell and the 24 V supply are both sources.

THE ONE RULE

24 V goes on **FIRST** and off **LAST**. The cell goes on **LAST** and off **FIRST**.

The Arduino's 5 V pin is the board's 5 V rail — plugging in the Arduino is switching on 5 V.

Power up

#	Action	Notes
1	Seat both supply leads with the output off	Black to J3 first, then red to J2. No live plugging.
2	24 V supply output ON	Set 24.0 V, current limit ~100 mA. The board draws almost nothing here; the limit is a tripwire.
3	Arduino USB in — this is “5 V on”	USB alone is enough for ~30 mA of coils. The 9 V barrel jack is optional.
4	Power on both 617 electrometers	Any time before the script. Allow them to warm up.
5	Power on the EDFA100P	Interlock shorted. Output terminated or already aimed at the cell.
6	Connect the cell, dark	PV– to J3 first, then PV+ to J1. Nothing running; the board sits in OPEN by default, which keeps R1 in circuit.
7	Start the script	It enables the pump itself and illuminates the cell.

Power down

#	Action	Notes
1	Let the script finish, or Ctrl-C	The abort path kills the pump first, before anything else.
2	Confirm the pump is off	Check the EDFA front-panel ENABLE indicator before approaching the output.
3	Disconnect the cell	Darken it, then PV+ out of J1, then PV– out of J3.
4	Arduino USB out — 5 V off	
5	24 V supply output OFF	Then red out of J2, then black out of J3 if unplugging at all.
6	EDFA and 617s off	Any order.

WHY THE ORDER MATTERS

With 5 V up, K1 energised and 24 V absent, K1 shorts out R1 and the cell drives straight into the DigiPot's input clamp — about 80% of its ± 20 mA absolute maximum. That is the case that destroys a chip. D4 sits between the two rails and does *not* protect the PV input path.

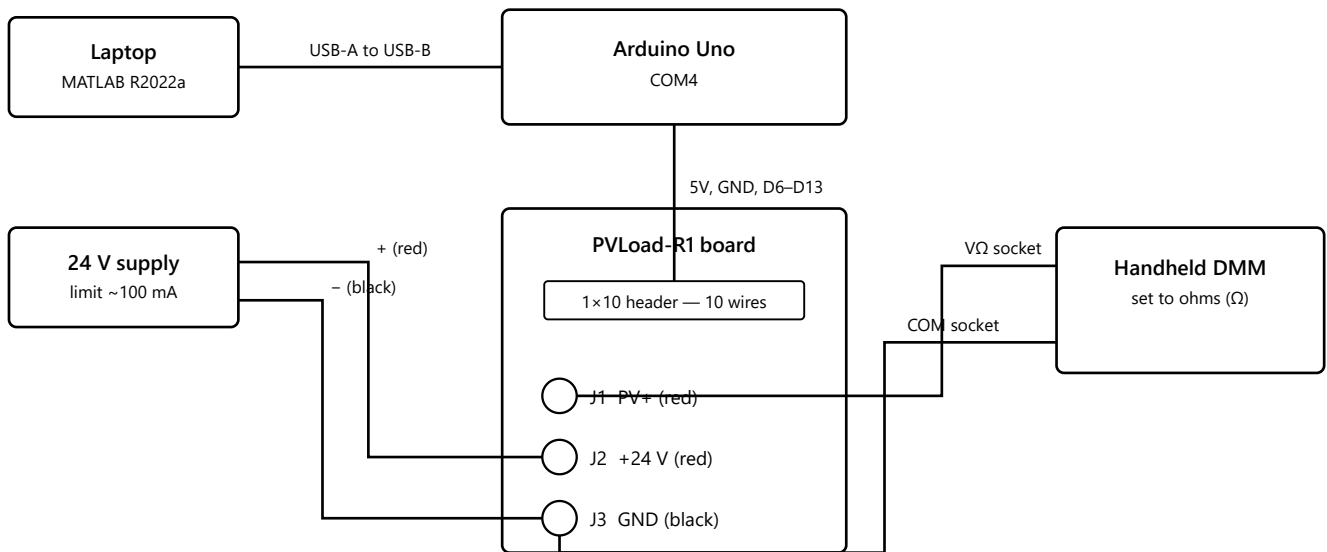
LASER

The EDFA100P is **CLASS 3B AT 1550 NM**, which is invisible. It emits up to 30 mW of amplified spontaneous emission with no optical input, at any pump current, whenever it is enabled. Terminate the output before running. Never look into the fibre bulkhead. If the interlock trips, it was opened by a person — software will not re-enable it and neither should you until you know why.

Shortcut for repeat runs: the jacks do not have to be cycled each time. A dark cell sources almost nothing, so blocking the light is equivalent to unplugging it and spares the connectors.

Wiring 1 · Handheld meter + 24 V

Board bring-up. No cell, no laser, no bench meters. The handheld stands in for the cell.



Both the supply return and the meter's COM lead land on J3. Use a stacking banana plug.

DO NOT put the red lead in the meter's 10A socket. In that socket the meter is a piece of wire and you would be shorting J1 to J3.

Software

`EDFA_ENABLED = false · DMM_ENABLED = false · RUN = "verify"` to check the board, or `"ramp"` to watch the whole range.

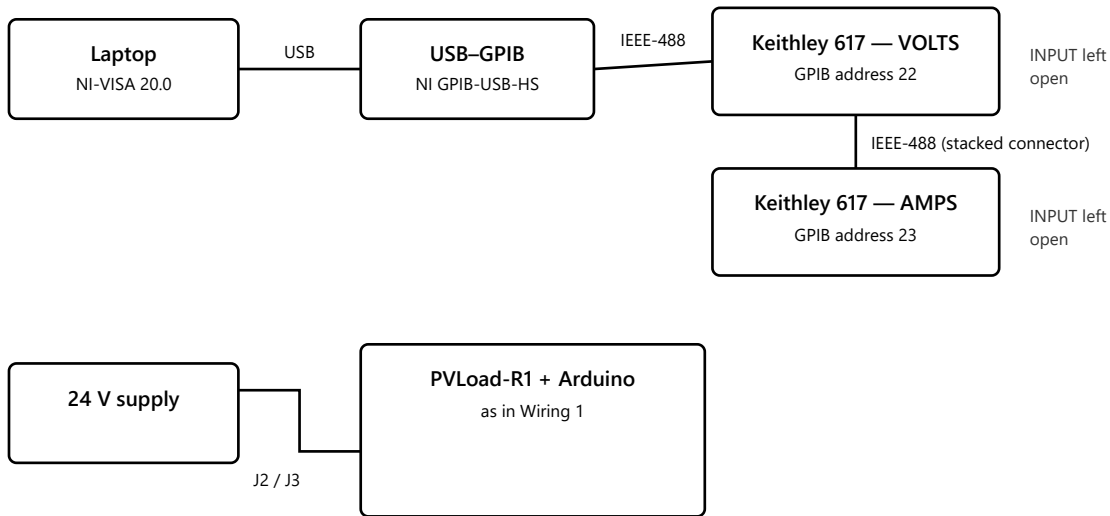
Power order for this setup

Up: 24 V output on → Arduino USB in → run. **Down:** stop → Arduino USB out → 24 V output off.

The meter leads stay plugged in throughout, both ways. In ohms mode a handheld is not a source and cannot hurt anything.

Wiring 2 • Two Keithley 617 + 24 V

Proving the GPIB chain and the 617 command dialect before a cell is involved. Meter inputs left open.



GPIB is a bus. One adapter serves both meters — IEEE-488 connectors stack, so the second cable runs meter to meter rather than back to the PC.

Before this works

- Give each 617 its own **primary address** from its front panel. They default to 27, which collides. The file expects 22 for volts and 23 for amps — change the meters or change the file.
- Confirm both appear: `visadevlist`. With no instruments attached it raises `unableToFindResourcesDefaultTimeout`, which is the healthy empty-bench answer. “Unable to find VISA **installations**” is the broken one and means NI-VISA is missing.
- `DMM_ENABLED = true`, `RUN = "meters"`.

What it proves

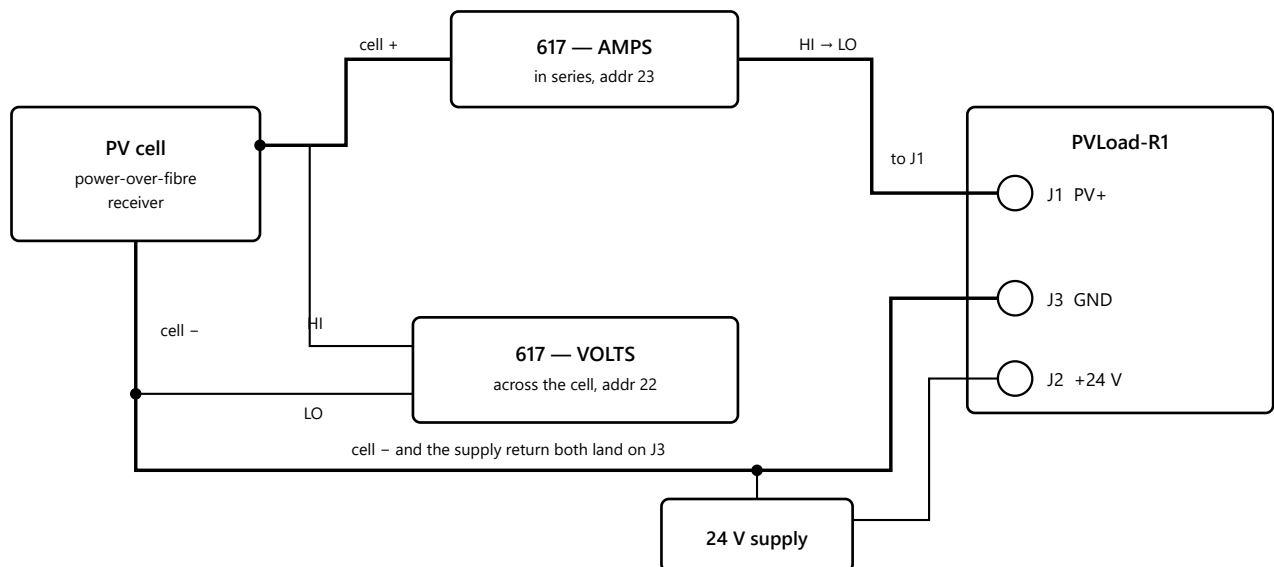
Both meters answer `U0` with a status word starting 617, accept their setup string, return a clean `U1` error word, and deliver ten readings. That covers the adapter, the cable, the addresses and the command dialect — everything except the measurement wiring, which Wiring 3 adds.

ABSOLUTE MAXIMUM ON THE 617 VOLTS INPUT IS 250 V PEAK. The amps function stops at 20 mA and there is nothing above it; a cell that outruns 20 mA needs an external shunt.

Wiring 3 · Full experiment

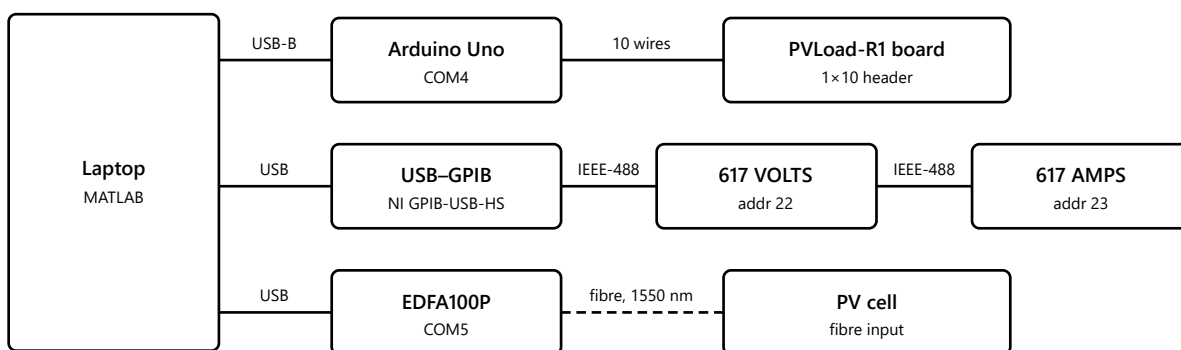
PV cell, both 617s, EDFA100P, 24 V supply, board. This is RUN = "sweep".

3a The measurement loop



Thick lines carry the cell current. The ammeter breaks the PV+ lead outside the board; the voltmeter taps the cell terminals directly, **upstream of the ammeter**.

3b Cables back to the PC



Three independent chains. GPIB is a bus, so both 617s hang off one adapter with the second cable running meter to meter.

THE VOLTMETER GOES ON THE CELL SIDE OF THE AMMETER — directly on the cell leads, not across J1 and J3. Downstream of the ammeter it reads low by the burden drop. On a 617 that is under 1 mV on every range except 20 mA, where it is 3 mV, so it no longer swamps the Isc end — but the rule is unchanged.

Software

RUN = "sweep" · EDFA_ENABLED = true · DMM_ENABLED = true · SERIAL_PORT, EDFA_PORT, DMM_V_ADDRESS, DMM_I_ADDRESS all set.

Enter the cell's real numbers in ISC_FULL, VOC_FULL and POWER_FULL — they size the ammeter range and print the estimate. An ISC_FULL above 20 mA is refused, because the 617 has no range above it.

For LEVEL_MODE = "power" the calibration arrays CAL_CURRENT_MA and CAL_POWER_MW must be filled from a measured pump-current-to-optical-power sweep. Outside their measured range the script refuses rather than extrapolating.

Output

pvload_<stamp>.csv for per-point readings and pvload_<stamp>_levels.csv for what is constant across a level, appended one level at a time so an abort keeps every completed level. Nothing is ever overwritten.